TEST YOUR UNDERSTANDING OF SECTION 32.5 In the standing wave described in Example 32.7, is there any point in the cavity where the energy density is zero at all times? If so, where? If not, why not?

ANSWER

no There are places where $\vec{E} = \mathbf{0}$ at all times (at the walls) and the electric energy density $\frac{1}{2}\epsilon_0 E^2$ is always zero. There are also places where $\vec{B} = \mathbf{0}$ at all times (on the plane midway between the walls) and the magnetic energy density $B^2/2\mu_0$ is always zero. However, there are *no* locations where both $\vec{E}$ and $\vec{B}$ are always zero. Hence the energy density at any point in the standing wave is always nonzero.

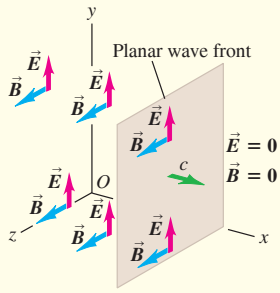
CHAPTER 32 SUMMARY

Maxwell's equations and electromagnetic waves: Maxwell's equations predict the existence of electromagnetic waves that propagate in vacuum at the speed of light, c. The electromagnetic spectrum covers frequencies from at least 1 to 10^{24} Hz and a correspondingly broad range of wavelengths. Visible light, with wavelengths from 380 to 750 nm, is a very small part of this spectrum. In a plane wave, $\vec{E}$ and $\vec{B}$ are uniform over any plane perpendicular to the propagation direction. Faraday's law and Ampere's law give relationships between the magnitudes of $\vec{E}$ and $\vec{B}$; requiring that both relationships are satisfied gives an expression for c in terms of ϵ_0 and μ_0. Electromagnetic waves are transverse; the $\vec{E}$ and $\vec{B}$ fields are perpendicular to the direction of propagation and to each other. The direction of propagation is the direction of $\vec{E} \times \vec{B}$.

$$E = cB \quad (32.4)$$

$$B = \epsilon_0\mu_0 cE \quad (32.8)$$

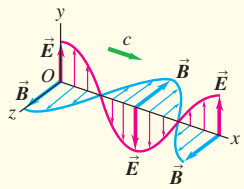
$$c = \frac{1}{\sqrt{\epsilon_0\mu_0}} \quad (32.9)$$



Sinusoidal electromagnetic waves: Equations (32.17) and (32.18) describe a sinusoidal plane electromagnetic wave traveling in vacuum in the $+x$-direction. If the wave is propagating in the $-x$-direction, replace $kx - \omega t$ by $kx + \omega t$. (See Example 32.1.)

$$\vec{E}(x, t) = \hat{\jmath}E_{max}\cos(kx - \omega t)$$
$$\vec{B}(x, t) = \hat{k}B_{max}\cos(kx - \omega t) \quad (32.17)$$

$$E_{max} = cB_{max} \quad (32.18)$$



Electromagnetic waves in matter: When an electromagnetic wave travels through a dielectric, the wave speed v is less than the speed of light in vacuum c. (See Example 32.2.)

$$v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{1}{\sqrt{KK_m}}\frac{1}{\sqrt{\epsilon_0\mu_0}} = \frac{c}{\sqrt{KK_m}} \quad (32.21)$$

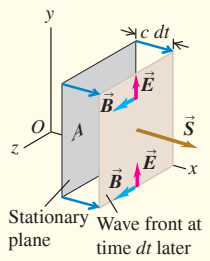
Energy and momentum in electromagnetic waves: The energy flow rate (power per unit area) in an electromagnetic wave in vacuum is given by the Poynting vector $\vec{S}$. The magnitude of the time-averaged value of the Poynting vector is called the intensity I of the wave. Electromagnetic waves also carry momentum. When an electromagnetic wave strikes a surface, it exerts a radiation pressure p_{rad}. If the surface is perpendicular to the wave propagation direction and is totally absorbing, $p_{rad} = I/c$; if the surface is a perfect reflector, $p_{rad} = 2I/c$. (See Examples 32.3–32.5.)

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B} \quad (32.28)$$

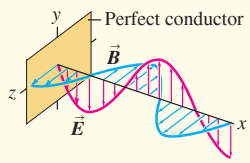
$$I = S_{av} = \frac{E_{max}B_{max}}{2\mu_0} = \frac{E_{max}^2}{2\mu_0 c} = \tfrac{1}{2}\sqrt{\frac{\epsilon_0}{\mu_0}}E_{max}^2 = \tfrac{1}{2}\epsilon_0 cE_{max}^2 \quad (32.29)$$

$$\frac{1}{A}\frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c} \quad (32.31)$$

(flow rate of electromagnetic momentum)



Standing electromagnetic waves: If a perfect reflecting surface is placed at $x = 0$, the incident and reflected waves form a standing wave. Nodal planes for $\vec{E}$ occur at $kx = 0, \pi, 2\pi, \ldots$, and nodal planes for $\vec{B}$ at $kx = \pi/2, 3\pi/2, 5\pi/2, \ldots$. At each point, the sinusoidal variations of $\vec{E}$ and $\vec{B}$ with time are 90° out of phase. (See Examples 32.6 and 32.7.)



Chapter 32 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 32.1 and 32.2 (Section 32.3) before attempting these problems.

VP32.2.1 A sinusoidal electromagnetic wave in vacuum has magnetic-field amplitude 4.30×10^{-3} T and wave number 2.50×10^6 rad/m. At a certain position and time the electric field $\vec{E}$ points in the $-y$-direction and the magnetic field $\vec{B}$ points in the $+x$-direction. Find (a) the amplitude of $\vec{E}$, (b) the wavelength, (c) the frequency, and (d) the direction of wave propagation.

VP32.2.2 The electric field in a sinusoidal electromagnetic wave in vacuum is given by $\vec{E}(y,t) = \hat{\imath}(2.45 \times 10^6\ \text{V/m}) \cos[(6.50 \times 10^6\,\text{rad/m})y - \omega t]$. Find (a) the wavelength, (b) the angular frequency ω, and (c) the direction of wave propagation. (d) Write the wave function for the magnetic field.

VP32.2.3 A dental laser emits a sinusoidal electromagnetic wave that propagates in vacuum in the positive x-direction with wavelength $2.94\ \mu$m. At $x = 0$ and $t = 0$ the electric field $\vec{E}$ points in the $+y$-direction and its magnitude equals the amplitude $E_{max} = 4.20$ MV/m. Find the magnitude and direction of (a) the magnetic field $\vec{B}$ at $x = 0$ and $t = 0$, (b) $\vec{E}$ at $x = 1.85\ \mu$m and $t = 0$, and (c) $\vec{B}$ at $x = 1.85\ \mu$m and $t = 0$.

VP32.2.4 The light from a red laser pointer has wavelength 6.35×10^{-7} m in vacuum. When this light is sent into a transparent material with relative permeability 1.00, its wavelength decreases to 3.47×10^{-7} m. Find (a) the frequency of the light in vacuum, (b) the frequency of the light in the material, (c) the speed of the light in the material, and (d) the dielectric constant of the material.

Be sure to review EXAMPLES 32.3 and 32.4 (Section 32.4) before attempting these problems.

VP32.4.1 A nonsinusoidal electromagnetic wave like that described in Section 32.2 has uniform electric and magnetic fields. The magnitude of the Poynting vector for this wave is 11.0 W/m^2. Find (a) the magnitudes of the electric and magnetic fields and (b) the energy density in the wave.

VP32.4.2 A sinusoidal electromagnetic wave in vacuum is given by the wave functions in Eqs. (32.17). Find the Poynting vector at (a) $x = 0$, $t = 0$; (b) $x = \lambda/4$, $t = 0$; (c) $x = \lambda/4$, $t = \pi/4\omega$.

VP32.4.3 A transmitter on the earth's surface radiates sinusoidal radio waves equally in all directions above the ground. An airplane flying directly over the radio station at an altitude of 12.5 km measures the wave from the station to have electric-field amplitude 0.360 V/m. Find (a) the magnetic-field amplitude and intensity measured by the airplane and (b) the total power emitted by the transmitter.

VP32.4.4 The sun emits light (which we can regard as a sinusoidal wave) equally in all directions. The distance from the sun to the earth is 1.50×10^{11} m, and the intensity of the sunlight that reaches the top of earth's atmosphere is 1.36 kW/m^2. Find (a) the amplitudes of the electric field and magnetic field at the top of the atmosphere and (b) the total power radiated by the sun.

Be sure to review EXAMPLES 32.6 and 32.7 (Section 32.5) before attempting these problems.

VP32.7.1 For the electromagnetic standing wave represented by Eqs. (32.34) and (32.35), find (a) the maximum magnitude of the Poynting vector anywhere in the standing wave; (b) the Poynting vector at $x = \lambda/8$, $t = 0$; (c) the Poynting vector at $x = \lambda/8$, $t = \pi/3\omega$; (d) the Poynting vector at $x = \lambda/8$, $t = 3\pi/4\omega$.

VP32.7.2 In a linearly polarized electromagnetic standing wave like that shown in Fig. 32.22, the amplitude of the magnetic field in the plane of the conductor is 1.20×10^{-7} T. The nodal plane of the magnetic field closest to the conductor is 3.60 mm from the conductor. Find (a) the wavelength, (b) the frequency, and (c) the amplitude of the electric field in the first nodal plane of the magnetic field.

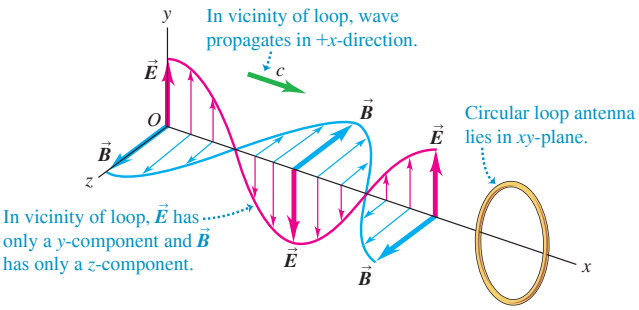
VP32.7.3 You set up electromagnetic standing waves in a cavity that has two parallel conducting walls, one at $x = 0$ and one at $x = 4.56$ cm. Find (a) the three lowest standing-wave frequencies and their corresponding wavelengths, and (b) the positions of the nodal planes of the electric field for each of these frequencies.

VP32.7.4 A microwave oven sets up a standing wave of wavelength 12.2 cm between two parallel conducting walls 48.8 cm apart. Find (a) the wave frequency and (b) the number of antinodal planes of the electric field between the walls.

BRIDGING PROBLEM Detecting Electromagnetic Waves

A circular loop of wire can be used as a radio antenna. If an 18.0-cm-diameter antenna is located 2.50 km from a 95.0 MHz source with a total power of 55.0 kW, what is the maximum emf induced in the loop? The orientation of the antenna loop and the polarization of the wave are as shown in **Fig. 32.24**. Assume that the source radiates uniformly in all directions.

Figure 32.24 Using a circular loop antenna to detect radio waves.



SOLUTION GUIDE

IDENTIFY and SET UP

1. The plane of the antenna loop is perpendicular to the direction of the wave's oscillating magnetic field. This causes a magnetic flux through the loop that varies sinusoidally with time. By Faraday's law, this produces an emf equal in magnitude to the rate of change of the flux. The target variable is the magnitude of this emf.

2. Select the equations that you'll need to find (i) the intensity of the wave at the position of the loop, a distance $r = 2.50$ km from the source of power $P = 55.0$ kW; (ii) the amplitude of the sinusoidally varying magnetic field at that position; (iii) the magnetic flux through the loop as a function of time; and (iv) the emf produced by the flux.

EXECUTE

3. Find the wave intensity at the position of the loop.
4. Use your result from step 3 to write expressions for the time-dependent magnetic field at this position and the time-dependent magnetic flux through the loop.
5. Use the results of step 4 to find the time-dependent induced emf in the loop. The amplitude of this emf is your target variable.

EVALUATE

6. Is the induced emf large enough to detect? (If it is, a receiver connected to this antenna will pick up signals from the source.)

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q32.1 By measuring the electric and magnetic fields at a point in space where there is an electromagnetic wave, can you determine the direction from which the wave came? Explain.

Q32.2 When travelling on a train on the upper level of the High Level Bridge in Newcastle you can easily pick up a number of radio stations on your pocket radio. But when jogging on the lower level of the bridge, which has steel girders on either side to support the upper level, the radio reception is much worse. Why is there a difference?

Q32.3 Give several examples of electromagnetic waves that are encountered in everyday life. How are they all alike? How do they differ?

Q32.4 Sometimes neon signs located near a powerful radio station are seen to glow faintly at night, even though they are not turned on. What is happening?

Q32.5 Is polarization a property of all electromagnetic waves, or is it unique to visible light? Can sound waves be polarized? What fundamental distinction in wave properties is involved? Explain.

Q32.6 Suppose that a positive point charge q is initially at rest on the x-axis, in the path of the electromagnetic plane wave described in Section 32.2. Will the charge move after the wave front reaches it? If not, why not? If the charge does move, describe its motion qualitatively. (Remember that $\vec{E}$ and $\vec{B}$ have the same value at all points behind the wave front.)

Q32.7 The light beam from a searchlight may have an electric-field magnitude of 1000 V/m, corresponding to a potential difference of 1500 V between the head and feet of a 1.5-m-tall person on whom the light shines. Does this cause the person to feel a strong electric shock? Why or why not?

Q32.8 For a certain sinusoidal wave of intensity I, the amplitude of the magnetic field is B. What would be the amplitude (in terms of B) in a similar wave of twice the intensity?

Q32.9 The magnetic-field amplitude of the electromagnetic wave from the laser described in Example 32.1 (Section 32.3) is about 100 times greater than the earth's magnetic field. If you illuminate a compass with the light from this laser, would you expect the compass to deflect? Why or why not?

Q32.10 Most automobiles have vertical antennas for receiving radio broadcasts. Explain what this tells you about the direction of polarization of $\vec{E}$ in the radio waves used in broadcasting.

Q32.11 If a light beam carries momentum, should a person holding a flashlight feel a recoil analogous to the recoil of a rifle when it is fired? Why is this recoil not actually observed?

Q32.12 A light source radiates a sinusoidal electromagnetic wave uniformly in all directions. This wave exerts an average pressure p on a perfectly reflecting surface a distance R away from it. What average pressure (in terms of p) would this wave exert on a perfectly absorbing surface that was twice as far from the source?

Q32.13 Does an electromagnetic *standing* wave have energy? Does it have momentum? Are your answers to these questions the same as for a *traveling* wave? Why or why not?

EXERCISES

Section 32.2 Plane Electromagnetic Waves and the Speed of Light

32.1 • (a) How much time does it take light to travel from the moon to the earth, a distance of 384,000 km? (b) Light from the star Sirius takes 8.61 years to reach the earth. What is the distance from earth to Sirius in kilometers?

32.2 • Consider each of the electric- and magnetic-field orientations given next. In each case, what is the direction of propagation of the wave? (a) $\vec{E}$ in the $+x$-direction, $\vec{B}$ in the $+y$-direction; (b) $\vec{E}$ in the $-y$-direction, $\vec{B}$ in the $+x$-direction; (c) $\vec{E}$ in the $+z$-direction, $\vec{B}$ in the $-x$-direction; (d) $\vec{E}$ in the $+y$-direction, $\vec{B}$ in the $-z$-direction.

32.3 • A sinusoidal electromagnetic wave is propagating in vacuum in the $+z$-direction. If at a particular instant and at a certain point in space the electric field is in the $+x$-direction and has magnitude 4.40 V/m, what are the magnitude and direction of the magnetic field of the wave at this same point in space and instant in time?

32.4 • Consider each of the following electric- and magnetic-field orientations. In each case, what is the direction of propagation of the wave? (a) $\vec{E} = E\hat{\imath}, \vec{B} = -B\hat{\jmath}$; (b) $\vec{E} = E\hat{\jmath}, \vec{B} = B\hat{\imath}$; (c) $\vec{E} = -E\hat{k}, \vec{B} = -B\hat{\imath}$; (d) $\vec{E} = E\hat{\imath}, \vec{B} = -B\hat{k}$.

Section 32.3 Sinusoidal Electromagnetic Waves

32.5 • **BIO Medical X Rays.** Medical x rays are taken with electromagnetic waves having a wavelength of around 0.10 nm in air. What are the frequency, period, and wave number of such waves?

32.6 • **BIO Ultraviolet Radiation.** There are two categories of ultraviolet light. Ultraviolet A (UVA) has a wavelength ranging from 320 nm to 400 nm. It is necessary for the production of vitamin D. UVB, with a wavelength in vacuum between 280 nm and 320 nm, is more dangerous because it is much more likely to cause skin cancer. (a) Find the frequency ranges of UVA and UVB. (b) What are the ranges of the wave numbers for UVA and UVB?

32.7 • Consider electromagnetic waves propagating in air. (a) Determine the frequency of a wave with a wavelength of (i) 5.9 km, (ii) 6.0 μm, (iii) 6.0 nm. (b) What is the wavelength (in meters and nanometers) of (i) gamma rays of frequency 6.47×10^{21} Hz and (ii) an AM station radio wave of frequency 590 kHz?

32.8 • An electromagnetic wave of wavelength 435 nm is traveling in vacuum in the $-z$-direction. The electric field has amplitude 2.70×10^{-3} V/m and is parallel to the x-axis. What are (a) the frequency and (b) the magnetic-field amplitude? (c) Write the vector equations for $\vec{E}(z, t)$ and $\vec{B}(z, t)$.

32.9 • An electromagnetic wave has an electric field given by $\vec{E}(y, t) = (3.60 \times 10^5 \text{ V/m})\hat{k}\cos[ky - (10.50 \times 10^{12} \text{ rad/s})t]$. (a) In which direction is the wave traveling? (b) What is the wavelength of the wave? (c) Write the vector equation for $\vec{B}(y, t)$.

32.10 • The electric field of a sinusoidal electromagnetic wave obeys the equation $E = (375 \text{ V/m}) \cos[(1.99 \times 10^7 \text{ rad/m})x + (5.97 \times 10^{15} \text{ rad/s})t]$. (a) What is the speed of the wave? (b) What are the amplitudes of the electric and magnetic fields of this wave? (c) What are the frequency, wavelength, and period of the wave? Is this light visible to humans?

32.11 • BBC World Service broadcasts at a frequency of 198 kHz. At a point some distance from the transmitter, the magnetic-field amplitude of the electromagnetic wave from BBC World Service is 4.82×10^{-11} T. Calculate (a) the wavelength; (b) the wave number; (c) the angular frequency; (d) the electric-field amplitude.

32.12 • An electromagnetic wave with frequency 65.0 Hz travels in an insulating magnetic material that has dielectric constant 3.64 and relative permeability 5.18 at this frequency. The electric field has amplitude 7.20×10^{-3} V/m. (a) What is the speed of propagation of the wave? (b) What is the wavelength of the wave? (c) What is the amplitude of the magnetic field?

32.13 • An electromagnetic wave with frequency 6.80×10^{14} Hz propagates with a speed of 2.35×10^8 m/s in a certain piece of glass. Find (a) the wavelength of the wave in the glass; (b) the wavelength of a wave of the same frequency propagating in air; (c) the index of refraction n of the glass for an electromagnetic wave with this frequency; (d) the dielectric constant for glass at this frequency, assuming that the relative permeability is unity.

Section 32.4 Energy and Momentum in Electromagnetic Waves

32.14 • BIO **High-Energy Cancer Treatment.** Scientists are working on a new technique to kill cancer cells by zapping them with ultrahigh-energy (in the range of 10^{12} W) pulses of light that last for an extremely short time (a few nanoseconds). These short pulses scramble the interior of a cell without causing it to explode, as long pulses would do. We can model a typical such cell as a disk 5.0 μm in diameter, with the pulse lasting for 4.0 ns with an average power of 2.0×10^{12} W. We shall assume that the energy is spread uniformly over the faces of 100 cells for each pulse. (a) How much energy is given to the cell during this pulse? (b) What is the intensity (in W/m^2) delivered to the cell? (c) What are the maximum values of the electric and magnetic fields in the pulse?

32.15 •• **Fields from a Light Bulb.** We can reasonably model a 90 W incandescent light bulb as a sphere 5.8 cm in diameter. Typically, only about 5% of the energy goes to visible light; the rest goes largely to nonvisible infrared radiation. (a) What is the visible-light intensity (in W/m^2) at the surface of the bulb? (b) What are the amplitudes of the electric and magnetic fields at this surface, for a sinusoidal wave with this intensity?

32.16 •• A sinusoidal electromagnetic wave from a radio station passes perpendicularly through an open window that has area 0.550 m^2. At the window, the electric field of the wave has rms value 0.0150 V/m. How much energy does this wave carry through the window during a 30.0 s commercial?

32.17 • A space probe 1.8×10^{10} m from a star measures the total intensity of electromagnetic radiation from the star to be 5.4×10^3 W/m^2. If the star radiates uniformly in all directions, what is its total average power output?

32.18 •• The energy flow to the earth from sunlight is about 1.4 kW/m^2. (a) Find the maximum values of the electric and magnetic fields for a sinusoidal wave of this intensity. (b) The distance from the earth to the sun is about 1.5×10^{11} m. Find the total power radiated by the sun.

32.19 •• A point source emits monochromatic electromagnetic waves into air uniformly in all directions. You measure the amplitude E_{max} of the electric field at several distances from the source. After graphing your results as E_{max} versus $1/r$, you find that the data lie close to a straight line that has slope 75.0 N · m/C. What is the average power output of the source?

32.20 • A sinusoidal electromagnetic wave emitted by a cellular phone has a wavelength of 34.1 cm and an electric-field amplitude of 5.00×10^{-2} V/m at a distance of 210 m from the phone. Calculate (a) the frequency of the wave; (b) the magnetic-field amplitude; (c) the intensity of the wave.

32.21 • A monochromatic light source with power output 52.0 W radiates light of wavelength 740 nm uniformly in all directions. Calculate E_{max} and B_{max} for the 740 nm light at a distance of 4.30 m from the source.

32.22 • **Television Broadcasting.** Sandy Heath transmitting station in Bedfordshire, England broadcasts a sinusoidal radio signal at a power of 180 kW. Assume that the wave spreads out uniformly into a hemisphere above the ground. At a home 5.00 km away from the antenna, (a) what average pressure does this wave exert on a totally reflecting surface, (b) what are the amplitudes of the electric and magnetic fields of the wave, and (c) what is the average density of the energy this wave carries? (d) For the energy density in part (c), what percentage is due to the electric field and what percentage is due to the magnetic field?

32.23 •• BIO **Laser Safety.** If the eye receives an average intensity greater than 1.0×10^2 W/m^2, damage to the retina can occur. This quantity is called the *damage threshold* of the retina. (a) What is the largest average power (in mW) that a laser beam 1.5 mm in diameter can have and still be considered safe to view head-on? (b) What are the maximum values of the electric and magnetic fields for the beam in part (a)? (c) How much energy would the beam in part (a) deliver per second to the retina? (d) Express the damage threshold in W/cm^2.

32.24 • In the Large Space Simulator facility at ESA's Test Centre in Noordwijk in the Netherlands, an array of powerful xenon lamps can produce a 6 m-diameter horizontal light beam of intensity 2700 W/m^2. (This simulates the intensity of sunlight on the planet Venus.) Find the average radiation pressure (in pascals and in atmospheres) on (a) a totally absorbing test surface and (b) a totally reflecting test surface. (c) Find the average momentum density (momentum per unit volume) in the light on the test surface.

32.25 • **Laboratory Lasers.** He–Ne lasers are often used in physics demonstrations. They produce light of wavelength 633 nm and a power of 0.500 mW spread over a cylindrical beam 1.00 mm in diameter (although these quantities can vary). (a) What is the intensity of this laser beam? (b) What are the maximum values of the electric and magnetic fields? (c) What is the average energy density in the laser beam?

32.26 • CP A totally reflecting disk has radius 8.00 μm and average density 600 kg/m^3. A laser has an average power output P_{av} spread uniformly over a cylindrical beam of radius 2.00 mm. When the laser beam shines upward on the disk in a direction perpendicular to its flat surface, the radiation pressure produces a force equal to the weight of the disk. (a) What value of P_{av} is required? (b) What average laser power is required if the radius of the disk is doubled?

Section 32.5 Standing Electromagnetic Waves

32.27 • **Microwave Oven.** The microwaves in a certain microwave oven have a wavelength of 12.2 cm. (a) How wide must this oven be so that it will contain five antinodal planes of the electric field along its width in the standing-wave pattern? (b) What is the frequency of these microwaves? (c) Suppose a manufacturing error occurred and the oven was made 5.0 cm longer than specified in part (a). In this case, what would have to be the frequency of the microwaves for there still to be five antinodal planes of the electric field along the width of the oven?

32.28 • An electromagnetic standing wave in air has frequency 85.5 MHz. (a) What is the distance between nodal planes of the $\vec{E}$ field? (b) What is the distance between a nodal plane of $\vec{E}$ and the closest nodal plane of $\vec{B}$?

32.29 •• An air-filled cavity for producing electromagnetic standing waves has two parallel, highly conducting walls separated by a distance L. One standing-wave pattern in the cavity produces nodal planes of the electric field with a spacing of 1.50 cm. The next-higher-frequency standing wave in the cavity produces nodal planes with a spacing of 1.25 cm. What is the distance L between the walls of the cavity?

PROBLEMS

32.30 •• **CALC** Consider a sinusoidal electromagnetic wave with fields $\vec{E} = E_{max}\hat{j}\cos(kx - \omega t)$ and $\vec{B} = B_{max}\hat{k}\cos(kx - \omega t + \phi)$, with $-\pi \leq \phi \leq \pi$. Show that if $\vec{E}$ and $\vec{B}$ are to satisfy Eqs. (32.12) and (32.14), then $E_{max} = cB_{max}$ and $\phi = 0$. (The result $\phi = 0$ means the $\vec{E}$ and $\vec{B}$ fields oscillate in phase.)

32.31 •• **BIO** **Laser Surgery.** Very short pulses of high-intensity laser beams are used to repair detached portions of the retina of the eye. The brief pulses of energy absorbed by the retina weld the detached portions back into place. In one such procedure, a laser beam has a wavelength of 810 nm and delivers 250 mW of power spread over a circular spot 510 μm in diameter. The vitreous humor (the transparent fluid that fills most of the eye) has an index of refraction of 1.34. (a) If the laser pulses are each 1.50 ms long, how much energy is delivered to the retina with each pulse? (b) What average pressure would the pulse of the laser beam exert at normal incidence on a surface in air if the beam is fully absorbed? (c) What are the wavelength and frequency of the laser light inside the vitreous humor of the eye? (d) What are the maximum values of the electric and magnetic fields in the laser beam?

32.32 •• **CP** (a) Estimate how long it takes to bring 1 cup (237 mL) of room-temperature water to 100°C in a microwave oven. (b) How much heat is required to heat this water? (c) Divide the required heat by the time in part (a) to estimate the power supplied to the water by the microwaves. (d) If this energy were supplied by a plane electromagnetic wave incident from one direction, and if half of the wave energy was absorbed by the water, then the average wave intensity would be the twice power estimate divided by the cross-sectional area of the cup seen by the wave. Use this model to estimate the average intensity of the microwaves in the oven. (e) Use Eqs. (32.29) to estimate the amplitude of the electric field in a microwave oven.

32.33 • A satellite 550 km above the earth's surface transmits sinusoidal electromagnetic waves of frequency 92.6 MHz uniformly in all directions, with a power of 25.0 kW. (a) What is the intensity of these waves as they reach a receiver at the surface of the earth directly below the satellite? (b) What are the amplitudes of the electric and magnetic fields at the receiver? (c) If the receiver has a totally absorbing panel measuring 20.0 cm by 40.0 cm oriented with its plane perpendicular to the direction the waves travel, what average force do these waves exert on the panel? Is this force large enough to cause significant effects?

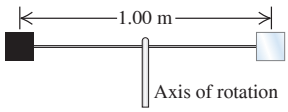
32.34 •• **CP** If we move a refrigerator magnet back and forth, we generate an electromagnetic wave that propagates away from us. Assume we move the magnet along the z-axis centered on the origin with an amplitude of 10 cm and with a frequency of 2.0 Hz. Consider a loop in the xy-plane centered on the origin with a radius of 2.0 cm. Assume that when the magnet is closest to the loop, the magnetic field within the loop has a spatially uniform value of 0.010 T. When the magnet has moved 10 cm in the direction away from the loop, the field in the loop becomes negligible. (a) Calculate the average rate at which the magnetic flux through the loop changes in time during each half-cycle of the motion of the magnet. (b) Use Faraday's law to estimate the average magnitude of the induced electric field at points on the loop. (c) Use Eq. (32.28) to estimate the average intensity of the electromagnetic wave in the xy-plane as it propagates radially away from the z-axis a distance of 2.0 cm. (d) Assume the intensity has the same value along the surface of a cylinder centered on the z-axis that extends 5.0 cm above and 5.0 cm below the xy-plane, and is negligible above or below this. (We are crudely postulating that the wave is emitted perpendicular to the axis of the motion in a narrow band.) Use this assumption to estimate the total power dissipated by the wave.

32.35 • The sun emits energy in the form of electromagnetic waves at a rate of 3.9×10^{26} W. This energy is produced by nuclear reactions deep in the sun's interior. (a) Find the intensity of electromagnetic radiation and the radiation pressure on an absorbing object at the surface of the sun (radius $r = R = 6.96 \times 10^5$ km) and at $r = R/2$, in the sun's interior. Ignore any scattering of the waves as they move radially outward from the center of the sun. Compare to the values given in Section 32.4 for sunlight just before it enters the earth's atmosphere. (b) The gas pressure at the sun's surface is about 1.0×10^4 Pa; at $r = R/2$, the gas pressure is calculated from solar models to be about 4.7×10^{13} Pa. Comparing with your results in part (a), would you expect that radiation pressure is an important factor in determining the structure of the sun? Why or why not?

32.36 • A small helium–neon laser emits red visible light with a power of 4.10 mW in a beam of diameter 3.10 mm. (a) What are the amplitudes of the electric and magnetic fields of this light? (b) What are the average energy densities associated with the electric field and with the magnetic field? (c) What is the total energy contained in a 1.00 m length of the beam?

32.37 •• **CP** Two square reflectors, each 1.50 cm on a side and of mass 4.00 g, are located at opposite ends of a thin, extremely light, 1.00 m rod that can rotate without friction and in vacuum about an axle perpendicular to it through its center (**Fig. P32.37**). These reflectors are small enough to be treated as point masses in moment-of-inertia calculations. Both reflectors are illuminated on one face by a sinusoidal light wave having an electric field of amplitude 1.35 N/C that falls uniformly on both surfaces and always strikes them perpendicular to the plane of their surfaces. One reflector is covered with a perfectly absorbing coating, and the other is covered with a perfectly reflecting coating. What is the angular acceleration of this device?

Figure **P32.37**



32.38 •• A source of sinusoidal electromagnetic waves radiates uniformly in all directions. At a distance of 15.0 m from this source, the amplitude of the electric field is measured to be 2.0 N/C. What is the electric-field amplitude 30.0 cm from the source?

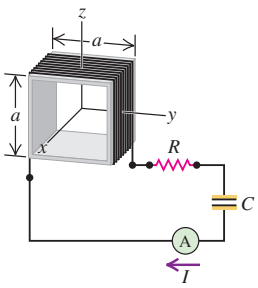
32.39 • **CP** **CALC** A cylindrical conductor with a circular cross section has a radius a and a resistivity ρ and carries a constant current I. (a) What are the magnitude and direction of the electric-field vector $\vec{E}$ at a point just inside the wire at a distance a from the axis? (b) What are the magnitude and direction of the magnetic-field vector $\vec{B}$ at the same point? (c) What are the magnitude and direction of the Poynting vector $\vec{S}$ at the same point? (The direction of $\vec{S}$ is the direction in which electromagnetic energy flows into or out of the conductor.) (d) Use the result in part (c) to find the rate of flow of energy into the volume occupied by a length l of the conductor. (*Hint:* Integrate $\vec{S}$ over the surface of this volume.) Compare your result to the rate of generation of thermal energy in the same volume. Discuss why the energy dissipated in a current-carrying conductor, due to its resistance, can be thought of as entering through the cylindrical sides of the conductor.

32.40 •• **CP** A circular wire loop has a radius of 7.60 cm. A sinusoidal electromagnetic plane wave traveling in air passes through the loop, with the direction of the magnetic field of the wave perpendicular to the plane of the loop. The intensity of the wave at the location of the loop is 0.0187 W/m^2, and the wavelength of the wave is 6.40 m. What is the maximum emf induced in the loop?

32.41 • In a certain experiment, a radio transmitter emits sinusoidal electromagnetic waves of frequency 110.0 MHz in opposite directions inside a narrow cavity with reflectors at both ends, causing a standing-wave pattern to occur. (a) How far apart are the nodal planes of the magnetic field? (b) If the standing-wave pattern is determined to be in its eighth harmonic, how long is the cavity?

32.42 ••• CP CALC An antenna is created by wrapping a square frame with side length a by a wire N times and then connecting the leads to a resistor R and a capacitor C, as shown in **Fig. P32.42**. The loop itself has self-inductance L. A plane electromagnetic wave with electric field $\vec{E} = E_{max}\cos(kz - \omega t)\hat{j}$ propagates in the $+z$-direction. The origin is at the center of the frame. (a) What is the magnetic flux through the coil in the direction of the $+x$-axis? (*Hint:* The identity $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$ may prove helpful.) (b) What is the emf generated in the coil? (c) The electromagnetic wave has frequency 4.00 Mz and intensity 100 W/m^2. The coil has $N = 50$ windings and side length $a = 10.0$ cm. It follows that its self-inductance is $L = 78.0\ \mu H$. If the resistance in the circuit is $R = 100\ \Omega$, what value of the capacitance C results in the resonance frequency of the L-R-C circuit being equal to the frequency of the wave? (d) What rms value of the current I_{rms} flows in that case?

Figure P32.42



32.43 •• CP **Global Positioning System (GPS).** The GPS network consists of 24 satellites, each of which makes two orbits around the earth per day. Each satellite transmits a 50.0 W (or even less) sinusoidal electromagnetic signal at two frequencies, one of which is 1575.42 MHz. Assume that a satellite transmits half of its power at each frequency and that the waves travel uniformly in a downward hemisphere. (a) What average intensity does a GPS receiver on the ground, directly below the satellite, receive? (*Hint:* First use Newton's laws to find the altitude of the satellite.) (b) What are the amplitudes of the electric and magnetic fields at the GPS receiver in part (a), and how long does it take the signal to reach the receiver? (c) If the receiver is a square panel 1.50 cm on a side that absorbs all of the beam, what average pressure does the signal exert on it? (d) What wavelength must the receiver be tuned to?

32.44 •• CP **Solar Sail.** NASA is giving serious consideration to the concept of *solar sailing*. A solar sailcraft uses a large, low-mass sail and the energy and momentum of sunlight for propulsion. (a) Should the sail be absorbing or reflective? Why? (b) The total power output of the sun is 3.9×10^{26} W. How large a sail is necessary to propel a 10,000 kg spacecraft against the gravitational force of the sun? Express your result in square kilometers. (c) Explain why your answer to part (b) is independent of the distance from the sun.

32.45 •• CP Interplanetary space contains many small particles referred to as *interplanetary dust*. Radiation pressure from the sun sets a lower limit on the size of such dust particles. To see the origin of this limit, consider a spherical dust particle of radius R and mass density ρ. (a) Write an expression for the gravitational force exerted on this particle by the sun (mass M) when the particle is a distance r from the sun. (b) Let L represent the luminosity of the sun, equal to the rate at which it emits energy in electromagnetic radiation. Find the force exerted on the (totally absorbing) particle due to solar radiation pressure, remembering that the intensity of the sun's radiation also depends on the distance r. The relevant area is the cross-sectional area of the particle, *not* the total surface area of the particle. As part of your answer, explain why this is so. (c) The mass density of a typical interplanetary dust particle is about 3000 kg/m^3. Find the particle radius R such that the gravitational and radiation forces acting on the particle are equal in magnitude. The luminosity of the sun is 3.9×10^{26} W. Does your answer depend on the distance of the particle from the sun? Why or why not? (d) Explain why dust particles with a radius less than that found in part (c) are unlikely to be found in the solar system. [*Hint:* Construct the ratio of the two force expressions found in parts (a) and (b).]

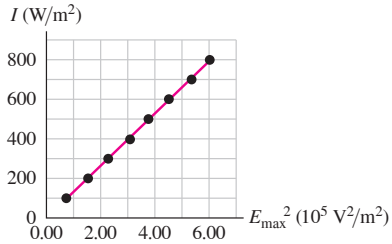
32.46 •• DATA The company where you work has obtained and stored five lasers in a supply room. You have been asked to determine the intensity of the electromagnetic radiation produced by each laser. The lasers are marked with specifications, but unfortunately different information is given for each laser:

Laser A: power = 2.6 W; diameter of cylindrical beam = 2.6 mm
Laser B: amplitude of electric field = 480 V/m
Laser C: amplitude of magnetic field = 8.7×10^{-6} T
Laser D: diameter of cylindrical beam = 1.8 mm; force on totally reflecting surface = 6.0×10^{-8} N
Laser E: average energy density in beam = 3.0×10^{-7} J/m^3

Calculate the intensity for each laser, and rank the lasers in order of increasing intensity. Assume that the laser beams have uniform intensity distributions over their cross sections.

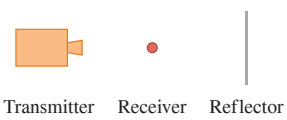
32.47 •• DATA Because the speed of light in vacuum (or air) has such a large value, it is very difficult to measure directly. To measure this speed, you conduct an experiment in which you measure the amplitude of the electric field in a laser beam as you change the intensity of the beam. **Figure P32.47** is a graph of the intensity I that you measured versus the square of the amplitude E_{max} of the electric field. The best-fit straight line for your data has a slope of 1.33×10^{-3} J/(V^2 · s). (a) Explain why the data points plotted this way lie close to a straight line. (b) Use this graph to calculate the speed of light in air.

Figure P32.47



32.48 •• DATA As a physics lab instructor, you conduct an experiment on standing waves of microwaves, similar to the standing waves produced in a microwave oven. A transmitter emits microwaves of frequency f. The waves are reflected by a flat metal reflector, and a receiver measures the waves' electric-field amplitude as a function of position in the standing-wave pattern that is produced between the transmitter and reflector (**Fig. P32.48**). You measure the distance d between points of maximum amplitude (antinodes) of the electric field as a function of the frequency of the waves emitted by the transmitter. You obtain the data given in the table.

Figure P32.48



f (10^9 Hz)	1.0	1.5	2.0	2.5	3.0	3.5	4.0	5.0	6.0	8.0
d (cm)	15.2	9.7	7.7	5.8	5.2	4.1	3.8	3.1	2.3	1.7

Use the data to calculate c, the speed of the electromagnetic waves in air. Because each measured value has some experimental error, plot the data in such a way that the data points will lie close to a straight line, and use the slope of that straight line to calculate c.

32.49 ••• **CP** When electromagnetic radiation strikes perpendicular to a flat surface, a totally absorbing surface feels radiation pressure I_0/c, where I_0 is the intensity of incident electromagnetic radiation. A totally reflecting surface feels twice that pressure. More generally, a surface absorbs a proportion e of the incident radiation and reflects a complementary proportion, $1 - e$, where e is the emissivity of the surface, as introduced in Chapter 17. Note that $0 \leq e \leq 1$. (a) Determine the radiation pressure p_{rad} in terms of I_0 and e. (b) Consider cosmic dust particles in outer space at a distance of 1.5×10^{11} m from the sun, where $I_{sun} = 1.4\ \text{kW/m}^2$. We can model these particles as tiny disks with $e = 0.61$, diameter 8.0 μm and mass 1.0×10^{-10} grams, all oriented perpendicular to the sun's rays. What is the force on one of these particles that is exerted by the radiation from the sun? (c) What is the ratio of this force to the attractive force of gravity exerted by the sun on the particle?

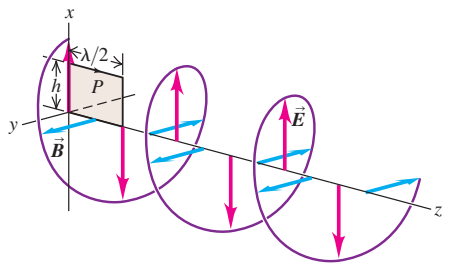
CHALLENGE PROBLEMS

32.50 ••• **CALC** An electromagnetic wave is specified by the following electric and magnetic fields:

$$\vec{E}(z, t) = E[\cos(kz - \omega t)\hat{\imath} + \sin(kz - \omega t)\hat{\jmath}]$$
$$\vec{B}(z, t) = B[-\sin(kz - \omega t)\hat{\imath} + \cos(kz - \omega t\hat{\jmath}]$$

where E and B are constant. (a) What is the value of the line integral $\oint \vec{E} \cdot d\vec{l}$ over the rectangular loop P shown in **Fig. P32.50**, which extends a distance h in the x-direction and half a wavelength in the z-direction, and passes through the origin? (b) What is the magnetic flux Φ_B through the loop P? (c) Use Faraday's law and the relationship $\omega/k = c$ to determine B in terms of E and c. (d) The wave specified above is "right circularly polarized." We can describe "left circularly polarized" light by swapping the signs on the two terms that involve sine functions. Superpose the above wave with such a reverse polarized analog. Write expressions for the electric and magnetic fields in that sum. Note that the result is a linearly polarized wave. (e) Consider right circularly polarized green light with wavelength 500 nm and intensity 100 W/m^2. What are the values of E and B? (*Hint:* Use Eq. (32.28) to determine the Poynting vector. Note that this is constant; its magnitude is the intensity of the light.)

Figure **P32.50**



32.51 ••• **CP** Electromagnetic radiation is emitted by accelerating charges. The rate at which energy is emitted from an accelerating charge that has charge q and acceleration a is given by

$$\frac{dE}{dt} = \frac{q^2a^2}{6\pi\epsilon_0 c^3}$$

where c is the speed of light. (a) Verify that this equation is dimensionally correct. (b) If a proton with a kinetic energy of 6.0 MeV is traveling in a particle accelerator in a circular orbit of radius 0.750 m, what fraction of its energy does it radiate per second? (c) Consider an electron orbiting with the same speed and radius. What fraction of its energy does it radiate per second?

32.52 ••• **CP** **The Classical Hydrogen Atom.** The electron in a hydrogen atom can be considered to be in a circular orbit with a radius of 0.0529 nm and a kinetic energy of 13.6 eV. If the electron behaved classically, how much energy would it radiate per second (see Challenge Problem 32.51)? What does this tell you about the use of classical physics in describing the atom?

32.53 ••• **CALC** Electromagnetic waves propagate much differently in *conductors* than they do in dielectrics or in vacuum. If the resistivity of the conductor is sufficiently low (that is, if it is a sufficiently good conductor), the oscillating electric field of the wave gives rise to an oscillating conduction current that is much larger than the displacement current. In this case, the wave equation for an electric field $\vec{E}(x, t) = E_y(x, t)\hat{\jmath}$ propagating in the $+x$-direction within a conductor is

$$\frac{\partial^2 E_y(x, t)}{\partial x^2} = \frac{\mu}{\rho}\frac{\partial E_y(x, t)}{\partial t}$$

where μ is the permeability of the conductor and ρ is its resistivity. (a) A solution to this wave equation is $E_y(x, t) = E_{max}e^{-k_C x}\cos(k_C x - \omega t)$, where $k_C = \sqrt{\omega\mu/2\rho}$. Verify this by substituting $E_y(x, t)$ into the above wave equation. (b) The exponential term shows that the electric field decreases in amplitude as it propagates. Explain why this happens. (*Hint:* The field does work to move charges within the conductor. The current of these moving charges causes i^2R heating within the conductor, raising its temperature. Where does the energy to do this come from?) (c) Show that the electric-field amplitude decreases by a factor of $1/e$ in a distance $1/k_C = \sqrt{2\rho/\omega\mu}$, and calculate this distance for a radio wave with frequency $f = 1.0$ MHz in copper (resistivity $1.72 \times 10^{-8}\ \Omega \cdot \text{m}$; permeability $\mu = \mu_0$). Since this distance is so short, electromagnetic waves of this frequency can hardly propagate at all into copper. Instead, they are reflected at the surface of the metal. This is why radio waves cannot penetrate through copper or other metals, and why radio reception is poor inside a metal structure.

MCAT-STYLE PASSAGE PROBLEMS

BIO **Safe Exposure to Electromagnetic Waves.** There have been many studies of the effects on humans of electromagnetic waves of various frequencies. Using these studies, the International Commission on Non-Ionizing Radiation Protection (ICNIRP) produced guidelines for limiting exposure to electromagnetic fields, with the goal of protecting against known adverse health effects. At frequencies of 1 Hz to 25 Hz, the maximum exposure level of electric-field amplitude E_{max} for the general public is 14 kV/m. (Different guidelines were created for people who have occupational exposure to radiation.) At frequencies of 25 Hz to 3 kHz, the corresponding E_{max} is $350/f$ V/m, where f is the frequency in kHz. (Source: "ICNIRP Statement on the 'Guidelines for Limiting Exposure to Time-Varying Electric, Magnetic, and Electromagnetic Fields (up to 300 GHz)'," 2009; *Health Physics* 97(3): 257–258.)

32.54 In most of the world, household electrical power is provided at a frequency of 50 Hz, so electromagnetic radiation at that frequency is of particular interest. On the basis of the ICNIRP guidelines, what is the maximum intensity of an electromagnetic wave at this frequency to which the general public should be exposed? (a) 7.7 W/m^2; (b) 160 W/m^2; (c) 65 kW/m^2; (d) 260 kW/m^2.

32.55 Doubling the frequency of a wave in the range of 25 Hz to 3 kHz represents what change in the maximum allowed electromagnetic-wave intensity? (a) A factor of 2; (b) a factor of $1/\sqrt{2}$; (c) a factor of $\frac{1}{2}$; (d) a factor of $\frac{1}{4}$.

32.56 The ICNIRP also has guidelines for magnetic-field exposure for the general public. In the frequency range of 25 Hz to 3 kHz, this guideline states that the maximum allowed magnetic-field amplitude is $5/f$ T, where f is the frequency in kHz. Which is a more stringent limit on allowable electromagnetic-wave intensity in this frequency range: the electric-field guideline or the magnetic-field guideline? (a) The magnetic-field guideline, because at a given frequency the allowed magnetic field is smaller than the allowed electric field. (b) The electric-field guideline, because at a given frequency the allowed intensity calculated from the electric-field guideline is smaller. (c) It depends on the particular frequency chosen (both guidelines are frequency dependent). (d) Neither—for any given frequency, the guidelines represent the same electromagnetic-wave intensity.

ANSWERS

Chapter Opening Question ?

(i) Metals are reflective because they are good conductors of electricity. When an electromagnetic wave strikes a conductor, the electric field of the wave sets up currents on the conductor surface that generate a reflected wave. For a perfect conductor, the requirement that the electric-field component parallel to the surface must be zero implies that this reflected wave is just as intense as the incident wave. Tarnished metals are less shiny because their surface is oxidized and less conductive; polishing the metal removes the oxide and exposes the conducting metal.

Key Example VARIATION Problems

VP32.2.1 (a) 1.29×10^6 V/m (b) 2.51×10^{-6} m (c) 1.19×10^{14} Hz (d) $+z$-direction

VP32.2.2 (a) 9.67×10^{-7} m (b) 1.95×10^{15} rad/s (c) $+y$-direction (d) $\vec{B}(y, t) = -\hat{k}(8.17 \times 10^{-3}\ \text{T})\cos[(6.50 \times 10^6\ \text{rad/m})y - (1.95 \times 10^{15}\ \text{rad/s})t]$

VP32.2.3 (a) 1.40×10^{-2} T, $+z$-direction (b) 2.89 MV/m, $-y$-direction (c) 9.64×10^{-3} T, $-z$-direction

VP32.2.4 (a) 4.72×10^{14} Hz (b) 4.72×10^{14} Hz (c) 1.64×10^8 m/s (d) 3.35

VP32.4.1 (a) $E = 64.4$ V/m, $B = 2.15 \times 10^{-7}$ T (b) 3.67×10^{-8} J/m^3

VP32.4.2 (a) $\hat{\imath}(E_{max}B_{max}/\mu_0)$ (b) zero (c) $\hat{\imath}(E_{max}B_{max}/2\mu_0)$

VP32.4.3 (a) $B_{max} = 1.20 \times 10^{-9}$ T, $I = 1.72 \times 10^{-4}$ W/m^2 (b) 1.69×10^5 W

VP32.4.4 (a) $E_{max} = 1.01 \times 10^3$ V/m, $B_{max} = 3.38 \times 10^{-6}$ T (b) 3.85×10^{26} W

VP32.7.1 (a) $E_{max}B_{max}/\mu_0$ (b) zero (c) $\hat{\imath}\sqrt{3}\,E_{max}B_{max}/2\mu_0$ (d) $-\hat{\imath}\,E_{max}B_{max}/\mu_0$

VP32.7.2 (a) 1.44 cm (b) 2.08×10^{10} Hz (c) 36.0 V/m

VP32.7.3 (a) $f_1 = 3.29 \times 10^9$ Hz, $\lambda_1 = 9.12$ cm; $f_2 = 6.58 \times 10^9$ Hz, $\lambda_2 = 4.56$ cm; $f_3 = 9.87 \times 10^9$ Hz, $\lambda_3 = 3.04$ cm (b) $x = 0$ and 4.56 cm; $x = 0$, 2.28 cm, and 4.56 cm; $x = 0$, 1.52 cm, 3.04 cm, and 4.56 cm

VP32.7.4 (a) 2.46×10^9 Hz (b) 8

Bridging Problem

0.0368 V